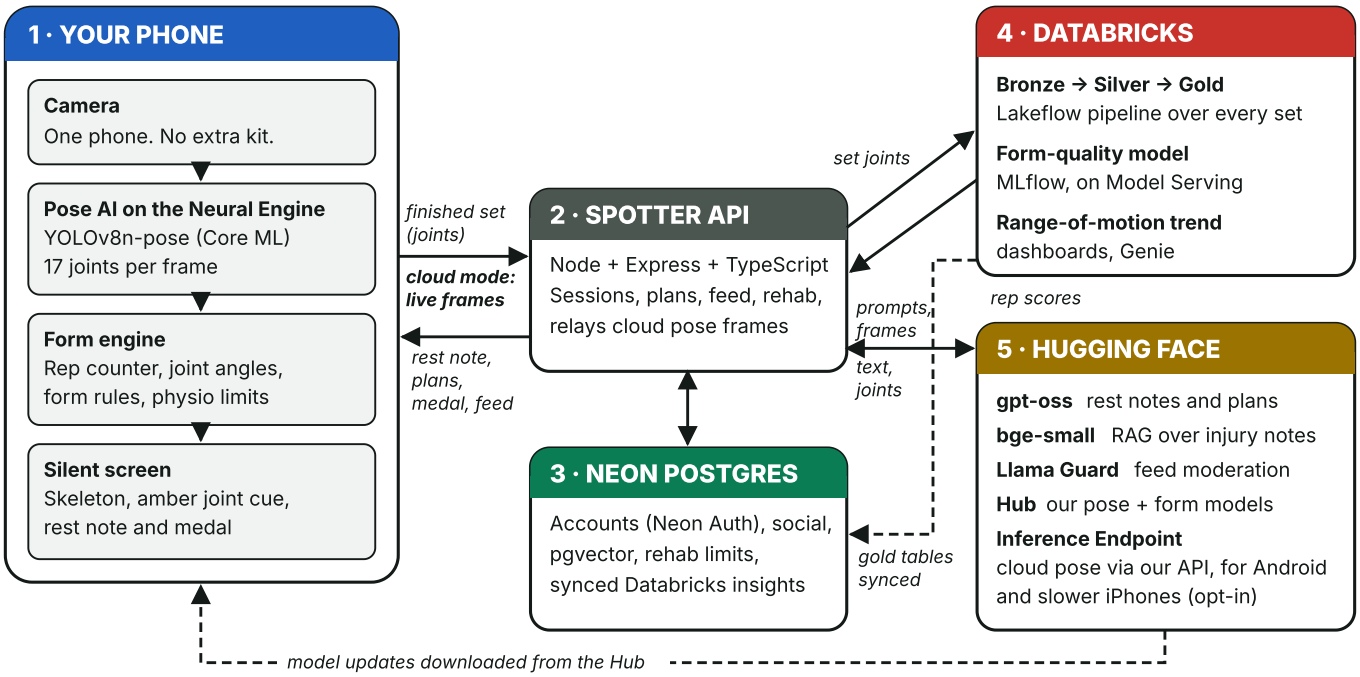
Spotter

A quiet coach for new gym-goers and people coming back from injury, on the phone you already own.

Pose AI runs on the phone when it can, and in the cloud only when it can't.

HOW IT FITS TOGETHER



THREE SPEEDS

DURING A REP · UNDER 0.1 S

Phone, or cloud pose

Reps and form are checked on the phone's chip, or by our cloud pose endpoint when the phone is too slow. Feedback is silent and visual.

DURING REST · ABOUT 1.5 S

Phone → Databricks → Hugging Face

Our model scores the set and writes one kind rest note. If the cloud is slow, the phone's rules set the medal.

OVER WEEKS

Plans and progress

New: a balanced weekly routine that works every muscle group.
Recovering: range of motion per joint, and a physio report.

BY THE NUMBERS

9.8 ms
pose AI per frame on an iPhone 17 Pro Max (p95 10.8 ms)

17
joints tracked per frame

12 fps
the bar a phone must clear to keep pose on-device; below it, cloud mode (with consent)

5
exercises today, each a JSON spec, so more are data, not code

WHAT EACH SPONSOR'S TECH DOES

Hugging Face

Open models end to end: our pose and form models on the Hub, gpt-oss via Inference Providers, bge-small retrieval, Llama Guard, and a pose Inference Endpoint that offloads phones too slow to run it.

Databricks

Every set flows through bronze, silver and gold. An MLflow model on Model Serving scores sets during rest. Gold tables sync to Neon, so the app never waits on a warehouse.

Neon

System of record: Neon Auth, the social graph, pgvector for retrieval and buddy matching, rehab limits and Databricks insights. Dev branches plus a seeded demo branch.

Safety lines we hold: no diagnosis, no treatment advice, never changes a physio's limits, and any mention of pain routes to "please check with your physio or a doctor". Cloud pose needs consent; frames are processed in memory and never stored. Deleting an account clears it from Neon and Databricks.
Code: github.com/Del0m/KnightHacks2026 · Knight Hacks 2026